

Ethan Villanueva

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SUMMARY

Final-year Computer Science student at the University of Nottingham (projected First-class honours) following a 13-month software engineering placement at Siemens Digital Industries. Experienced in the end-to-end development lifecycle within a large-scale C#/.NET enterprise environment, automated testing, and CI/CD pipeline management.

EDUCATION

University of Nottingham

Sep 2023 - Jun 2027

Bsc Computer Science with Year in Industry

- Projected First-class honours
- Relevant modules: Developing Maintainable Software, Databases and Interfaces, Operating Systems & Concurrency, Systems and Architecture, Artificial Intelligence Methods, Mathematics for Computer Scientists, Programming and Algorithms, Programming Paradigms, Software Engineering Group Project

King Edward VI College

Sep 2020 - Jun 2022

A-Levels

- Achieved AAA in Physics, Maths, Geology.

Queen Mary's Grammar School

Sep 2015 - Jun 2020

GCSEs

- Obtained All 7's/8's (A/A*) across 11 subjects including IT, Maths, Physics, and English.

TECHNICAL SKILLS

- **Languages:** C#, Python, Java, C, SQL, JavaScript, HTML/CSS
- **Frameworks/Libraries:** .NET, WPF/MVVM, Scikit-Learn, NumPy, Pandas, Matplotlib
- **Databases/Cloud:** SQL Server, PostgreSQL, AWS DynamoDB, SQLite, Supabase, AWS (EC2, Lambda)
- **DevOps/Tools:** Azure DevOps, Git, CI/CD Pipelines, YAML, PowerShell, Bash, NuGet
- **Testing/Quality:** FlaUI, Playwright, JUnit, TDD, Automated UI Testing and Regression
- **Systems/Architecture:** Multi-threading, POSIX Threads, TCP/IP, Linux/Unix, Berkeley Sockets

EXPERIENCE

Siemens Digital Industries Software

Jul 2025 - Aug 2026 | Milton Keynes

Software Engineer Intern

- Built and shipped product features for Simcenter FLOMASTER, tracing changes through WPF/MVVM UI, C# simulation logic, SQL Server stored procedures, and NuGet package dependencies across 7 repositories.
- Automated SDK documentation generation by engineering Azure DevOps YAML pipelines and PowerShell scripts, replacing legacy CHM formats with modern HTML and **eliminating all manual build steps per release**.
- Resolved a critical CI/CD bottleneck by tracing a severe UI-test VM slowdown to a SQL Server 2025 incompatibility, **cutting automated regression test runtime by ~50% (from 4 hours to 2 hours)** by reconfiguring the test environment on SQL Server 2022.
- Authored automated unit, integration, and UI tests for all new features and bug fixes, **expanding UI test suite by ~15%** and maintaining zero regression leaks across release cycles.
- Onboarded and trained incoming interns on C#, MVVM, FlaUI, and NuGet; conducted pull request reviews and delivered structured handover documentation.

Capital One (University Project sponsored by Capital One)

Oct 2024 - May 2025 | Remote

Back-end Developer

- Architected AWS DynamoDB schemas and mock REST APIs in an 8-person Agile team to simulate real-time banking transactions and notification dispatch for a Capital One Communication Channel.
- Researched and applied GDPR data privacy controls to ensure the system architecture adhered to financial data compliance requirements, documenting decisions as formal acceptance criteria.
- Collaborated with Capital One engineers across regular agile sprints, defining user stories, acceptance criteria, and API contracts to align the university-side implementation with industry requirements.

PROJECTS

Mindmaply - LeedsHack 2026 Winner: Best Use of Google Gemini API (Python, Gemini API, Vector Embeddings)

- Engineered the core Python conversation engine and graph data models to support non-linear LLM conversation branching across dynamically created topic nodes.
- Implemented intent detection using vector embeddings and Gemini API prompts to automatically branch conversations when subjects shift, without user intervention.
- Built context scoping logic and automated background summarisation, slashing prompt token overhead and preventing model drift across long sessions.
- Handled state orchestration and integrated backend graph traversal with the frontend UI.

Operating System Simulator (C, POSIX Threads, Linux/Unix)

- Built an optimised round-robin scheduler to manage multi-threaded task queues with fairness and efficiency across concurrent processes.
- Implemented POSIX threads, mutexes, and semaphores to handle concurrency and prevent race conditions.
- Verified reliability by eliminating deadlocks and memory leaks through rigorous testing with Valgrind.

Forest Fire Prediction Models (Python, Scikit-Learn, Pandas, Matplotlib)

- Built linear regression and neural network predictive models to predict forest fire burn areas from meteorological data.
- Evaluated models with k-fold cross-validation, achieving a **Mean Squared Error of 0.058** (normalised to [0,1]) with the neural network model.
- Visualised dataset insights, regression residuals, and model performance in Jupyter Notebook using Matplotlib.

Route Network Server (C, TCP/IP, Berkeley Sockets)

- Engineered a multi-client TCP/IP server to query and manage network graphs over persistent socket connections.
- Implemented Dijkstra's Algorithm over a custom graph library to calculate optimal transit paths in real time.
- Designed a custom client-server protocol supporting dynamic node and edge updates with persistent connection state.

Vehicle Management System (HTML, CSS, JavaScript, PostgreSQL, Supabase)

- Built a responsive full-stack web application managing vehicle and owner records via a Supabase PostgreSQL backend with structured relational tables.
- Implemented RESTful API interactions for user queries and automated end-to-end integration tests using Playwright.
- Audited performance and accessibility with Google Lighthouse.

Test-Driven Development Group Project (Java, JUnit)

- Led a team of 3 in refactoring a legacy property viewing application, applying red-green-refactor TDD cycles and writing comprehensive automated tests in JUnit throughout.
- Achieved **96.8% and 94.0% code coverage** across the two primary domain classes.
- Decoupled tightly linked legacy components into clean, modular classes following SOLID design principles.

CERTIFICATIONS/AWARDS

- **AWS Certified Cloud Practitioner** - Amazon Web Services
- **LeedsHack 2026 Winner: Best Use of Google Gemini API** - Major League Hacking